

Exponent Laws with Variable Bases

Teacher's Edition

The Period

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Must Whole-number exponents on monomial bases: multiply and divide monomials, and raise a monomial to a power, generating an equivalent expression each time.

Must not Whole-number exponents today; the negative ones are tomorrow. No rational exponents, and monomials only — nothing with a plus sign in it.

Target I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 2 and 7 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	13 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. Every law from last week is the same law. The base is a letter now; nothing else changed.
2. An exponent outside the parentheses reaches every factor inside — the coefficient included.
3. Only like bases combine. An x and a y never merge.

Slide by Slide

1. **Exponent Laws with Variable Bases.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the bridge: $y \cdot y \cdot y \cdot y \cdot y$ is y^5 for exactly the reason $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ is 2^5 .
3. **Warm-Up — Answers.** Reveal. 1. 4 2. $6^2 = 36$ 3. 7 and 8 4. y^5
4. **Notes — Copy both lines.** Coefficients multiply. Exponents add. Two different operations in the same problem.
5. **Notes.** Everything inside the parentheses gets the exponent — the number included.
6. **Notes.** x^7 over x^2 cancels two x's and leaves five: subtract.
7. **Example 1 — Multiplying two monomials.** SPOKEN ONLY.
8. **Example 1 — Worked.** Regroup by kind, then apply the product law twice.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the missing exponent 1 on a and the exponent-multiplier ($15a^2b^8$).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — A power of a product, then a quotient.** SPOKEN ONLY.
12. **Example 2 — Worked.** Raise every factor in the numerator: $3^3 = 27$, $(m^2)^3 = m^6$, n^3 .
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the un-raised coefficient ($2p^{12}/8p^5 = p^7/4$) and the exponent-adder — $2p^2$ if the 3 and 4 are added in the power step, $2p^{17}$ if the 12 and 5 are added in the quotient step.
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** $x^5 \cdot x^8 \rightarrow x^{13}$ x^{40} — multiplied the exponents [B1G-M 8.AR.1.1 misconception: misapplied laws]; $2x^{13}$ — an invented coefficient
- 17-18. **Board 2.** $(y^4)^3 \rightarrow y^{12}$ y^7 — added the exponents [8.AR.1.1: power of a power confused with a product]
- 19-20. **Board 3.** A cube has edges $2x^3$ centimeters long. Its surface area is $24x^6$ square centimeters. What is the volume of the cube, in cubic centimeters? $\rightarrow 8x^9$ The surface area is not needed. $2x^9$ — the coefficient was not raised [B1G-M 8.AR.1.1: power of a product]; $8x^6$ — exponents added instead of multiplied [confused with product of powers]; anything built from $24x^6$ — the unneeded figure was used
- 21-22. **Board 4.** $(3a^2b)(4ab^5) \rightarrow 12a^3b^6$ $7a^3b^6$ — coefficients added instead of multiplied [8.AR.1.1: coefficients are factors]; $12a^2b^5$ — the factors a and b with exponent 1 were dropped
- 23-24. **Board 5.** $(5m^3)^2$ Which expression is equivalent? $\rightarrow A - 25m^6$ (B) $5 \cdot 2$ instead of 5^2 — the exponent was multiplied into the coefficient [B1G-M 8.AR.1.1 misconception: multiplying the exponent into the base]; (C) $3 + 2$ instead of $3 \cdot 2$ [power of a power confused with a product of powers]; (D) the coefficient was not raised at all [8.AR.1.1: power of a product] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.
- 25-26. **Board 6.** $\frac{12x^8}{3x^2}$ Which expression is equivalent? $\rightarrow B - 4x^6$ (A) $8 + 2$ — added the exponents [quotient law confused with product law]; (C) $12 - 3$ — the

coefficients were subtracted instead of divided [8.AR.1.1: coefficients are factors]; (D) $8 \div 2$ — the exponents were divided instead of subtracted [B1G-M 8.AR.1.1: misapplied quotient law] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. $\frac{c^{10}d^3}{c^2d^3} \rightarrow c^8$ $c^8d - d^0$ read as d [8.AR.1.1: zero exponent]; 0 or $c^8 \cdot 0 - d^0$ read as 0 [8.AR.1.1: zero exponent, $d^0 = 1$]

29-30. Board 8. Find the value of n : $x^3 \cdot x^n = x^{11} \rightarrow n = 8$ $14 -$ added 3 to 11; $33 -$ multiplied [8.AR.1.1: product law is additive in the exponents]

31-32. Board 9. Rachel writes $(14a^3b)^3$ as $14 \cdot 14 \cdot 14 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b$. Justina writes it as $2,744a^9b^3$. Who is correct? Explain. $\rightarrow$ **Both are correct; $14^3 = 2,744$, nine a 's make a^9 , and $b \cdot b \cdot b = b^3$.** 'Justina only' — the expanded form is not recognized as equivalent [8.AR.1.1: generate equivalent expressions]; 'Rachel only' or ' $14a^9b^3$ ' — the coefficient was not cubed [B1G-M 8.AR.1.1: power of a product]; ' $42a^9b^3$ ' — the coefficient multiplied by 3 instead of cubed

33. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. x^{40} — multiplied the exponents [B1G-M 8.AR.1.1 misconception: misapplied laws]

Board 2. y^7 — added the exponents [8.AR.1.1: power of a power confused with a product]

Board 3. $2x^9$ — the coefficient was not raised [B1G-M 8.AR.1.1: power of a product]

Board 4. $7a^3b^6$ — coefficients added instead of multiplied [8.AR.1.1: coefficients are factors]

Board 5. $5 \cdot 2$ instead of 5^2 — the exponent was multiplied into the coefficient [B1G-M 8.AR.1.1 misconception: multiplying the exponent into the base]

Board 6. $8 + 2$ — added the exponents [quotient law confused with product law]

Board 7. $c^8d - d^0$ read as d [8.AR.1.1: zero exponent]

Board 8. $14 -$ added 3 to 11

Board 9. 'Justina only' — the expanded form is not recognized as equivalent [8.AR.1.1: generate equivalent expressions]

Materials Whiteboards and markers. Calculators are allowed on everything; there is little for them to do today beyond 14^3 on whiteboard 9.